



Physics-guided diffusion models for inverse design of disordered metamaterials



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Disordered metamaterials are promising for programming physical properties, yet their inverse design remains challenging due to the non-intuitive structure-property relationships. Recent generative approaches, particularly diffusion models, have shown potential in high-dimensional inverse design tasks. However, existing methods typically rely on task-specific training strategies, such as conditional data-driven or physics-informed loss functions. Consequently, models must be retrained from scratch whenever governing equations, boundary conditions, or design objectives change, limiting their flexibility and generalization. In this work, we propose physics-guided diffusion models that leverage differentiable physics-based solvers to instantly guide the generative process for inverse design. Drawing inspiration from classifier guidance, we develop a sampling strategy that directly incorporates physics guidance into the reverse stochastic differential equations. Using gradients from differentiable solvers, our approach enables task-adaptive generation within a learned morphology prior, while requiring the diffusion model to be trained only once on unlabeled data. Focusing on 2D disordered closed-cell foam metamaterials, we present three design tasks: (1) achieving target effective thermal conductivity, (2) matching desired load-displacement response, and (3) maximizing energy absorption involving fractures. The results in each scenario demonstrate the versatility, efficiency, and practicality of physics-guided diffusion models for tackling complex inverse design problems in disordered metamaterials and beyond.

Metamaterials have revolutionized the fields of mechanics, thermology, acoustics and optics by offering special material properties not found in nature^{1–5}. While periodic metamaterials have been extensively studied, disordered structures have recently attracted increasing attention due to their unique advantages, including enhanced defect and damage tolerance, greater robustness to fabrication variability, and increased multifunctionality and programmability^{6,7}. However, the inverse design of disordered metamaterials presents a formidable challenge. Unlike periodic structures where properties can be engineered through systematic modification of a few unit-cell parameters, disordered systems lack such clear design principles. The structure-property relationships are highly complex and non-intuitive, making designing disordered structures with targeted properties exceedingly difficult and rendering traditional trial-and-error approaches impractical.

Conventional inverse design methods for disordered metamaterials face considerable hurdles. Gradient-based approaches such as topology optimization (TO)⁸ are susceptible to local optima, particularly in disordered systems⁹. Additionally, gradient-based methods struggle to incorporate complex morphological priors, often producing free-form geometries that fail to match the required statistical or geometric characteristics. Gradient-free methods such as evolutionary algorithms¹⁰ and simulated annealing¹¹ offer better global exploration but become computationally prohibitive as the dimensionality of the design space grows.

Recent advances in generative models offer a paradigm shift for addressing these challenges in metamaterial inverse design. Generative models, including variational autoencoders (VAEs)¹², generative adversarial networks (GANs)¹³, flow models¹⁴ and diffusion models^{15,16}, learn the underlying distribution of the training data and can synthesize new samples

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that preserve the statistical characteristics of the original dataset. Unlike conventional inverse design methods, these models naturally encode morphological priors through training, ensuring the generated structures inherit realistic geometric features from existing designs. Moreover, generative models are inherently stochastic and can produce diverse candidate solutions, exploring the design space beyond local optima. In recent works, physics-informed diffusion models have shown great potential in inverse design tasks^{17–20}. However, most of the current generative models for inverse design require labeled data or physical residuals in the training process, which often incurs substantial labeling effort or additional training cost. Most importantly, such models are typically restricted to a single type of design problem; adaptation to new design problems (e.g., different governing equations, boundary conditions, design objectives, etc.) generally requires retraining the entire model, thereby limiting their flexibility and practicality.

Classifier guidance²¹ has emerged as an effective strategy to steer diffusion models toward desired categories or attributes for conditional generation. However, training an additional classifier on noisy intermediate data is often difficult. Herein, we propose physics-guided diffusion models that replace the classifier-based guidance term in the sampling process with physics-based guidance enabled by differentiable solvers^{22,23}. For demonstration, Fig. 1 shows an inverse design process in which physics guidance leads to a structure with a prescribed load-displacement response. The design starts from Gaussian noise and seeks to generate a physically consistent sample within 50 denoising steps. In an unconditional diffusion model, stochastic perturbations introduced during denoising may drive the generated structures away from the target response. Physics guidance is therefore applied to steer the generation toward the desired trajectory.

This framework brings three key advantages. First, the diffusion model can be trained on unlabeled datasets, substantially reducing the effort required for data preparation. Second, the optimization is driven by physics-based objectives rather than learned approximations, improving accuracy and reliability. Third, a well-trained model can be readily adapted to

multiple objectives without retraining the model. We validate the effectiveness and flexibility of the proposed method through three diverse numerical case studies on inverse design of 2D disordered closed-cell foam metamaterials, including (1) achieving target effective thermal conductivity, (2) matching desired load-displacement response, and (3) maximizing energy absorption involving fractures under a prescribed volume-fraction constraint. In this work, although we focus on the inverse design of materials within a learned foam morphology, the methodology can be applied to other engineering design problems through appropriate modifications of the priors, solvers and objectives.

Results

We present three inverse design tasks of 2D disordered closed-cell foams to demonstrate the proposed physics-guided diffusion models. All tasks use the same well-trained diffusion model to achieve the design of thermal conductivity, load-displacement response, and energy absorption, respectively. The detailed problem formulations can be found in Supplementary Section 1.

Design of effective thermal conductivity

We consider a simple steady-state heat conduction problem in this subsection, and the aim is to design foam structures with a target effective thermal conductivity. The computational domain Ω represents a region with dimension $1 \times 1 \text{ m}^2$ between two walls, containing a disordered foam structure and voids filled with gas, as illustrated in Fig. 2a. The top boundary is maintained at temperature $T_1 = 500 \text{ K}$, while the bottom boundary is held at $T_2 = 300 \text{ K}$. The left and right walls are assumed to be adiabatic. JAX-FEM²³, a differentiable GPU-accelerated solver based on finite element method (FEM), is used here to solve for the steady-state temperature field, and an example of steady-state solution is shown in Fig. 2b. Then, the effective thermal conductivity k_{eff} of the structure can be derived based on evaluation of the obtained temperature field²⁴.

To design disordered foams with target effective thermal conductivity, the sampling stochastic differential equation (SDE) of the trained diffusion

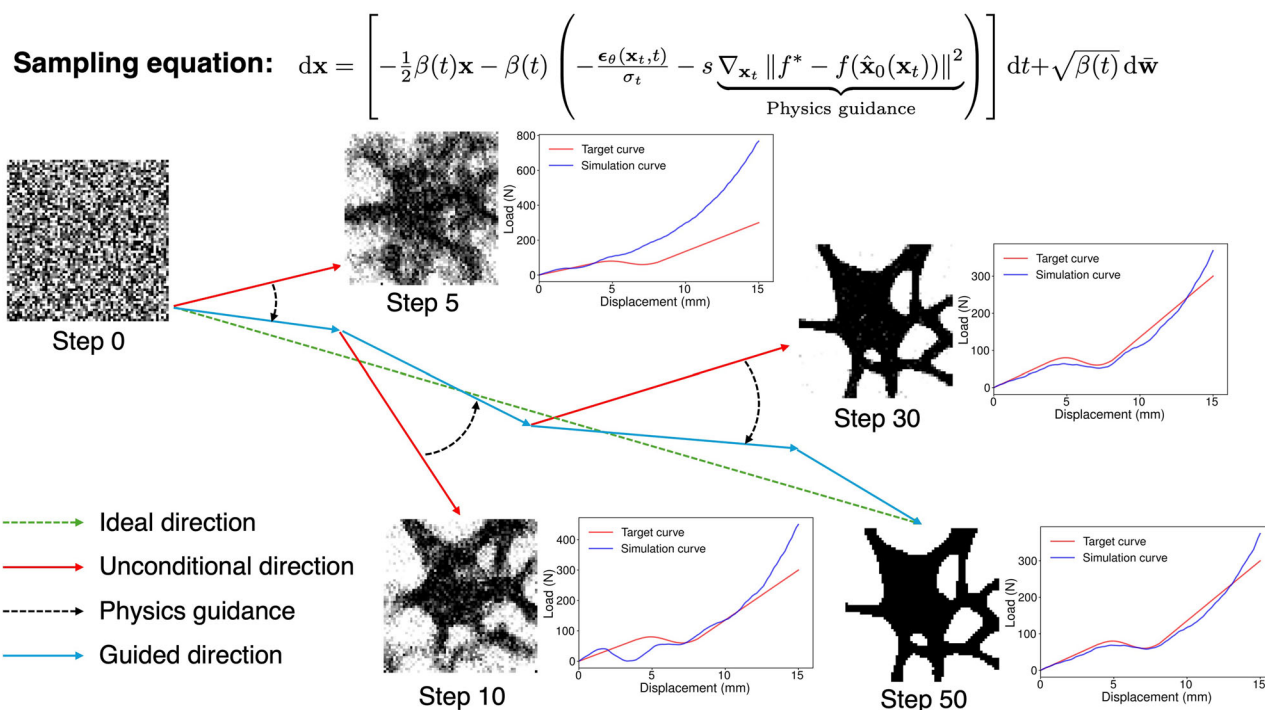


Fig. 1 | Illustration of physics-guided sampling process in diffusion models. To achieve a target load-displacement response, the design starts from Gaussian noise and seeks to generate a physically consistent sample within 50 denoising steps. In an

unconditional diffusion model, stochastic perturbations introduced during denoising may drive the generated structures away from the ideal path. Physics guidance is therefore applied to guide the generation toward the desired direction.

Fig. 2 | Formulation of the foam heat transfer problem. **a** Formulation of the foam heat transfer problem. The top and bottom boundary are maintained at constant temperature. The left and right walls are assumed to be adiabatic. **b** Solution of the steady-state temperature field.

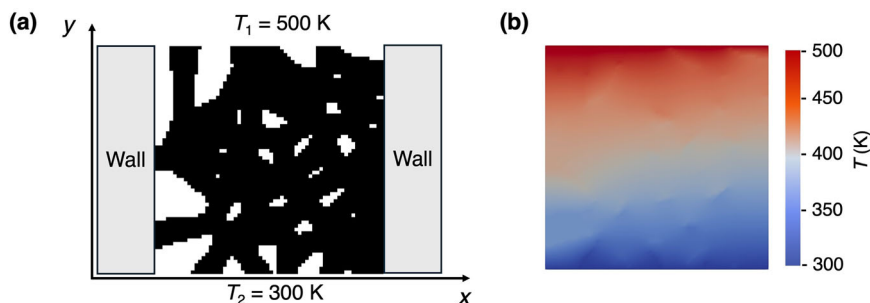
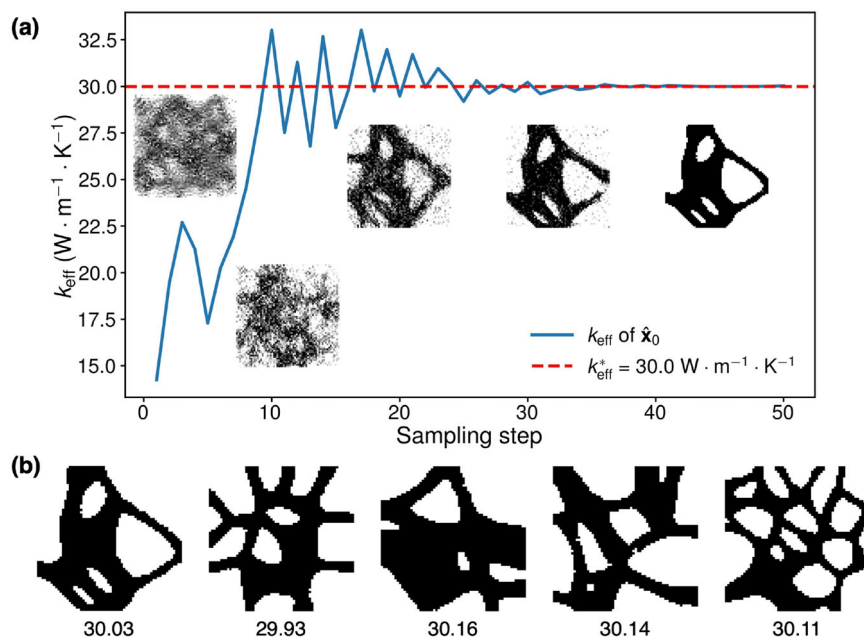


Fig. 3 | Inverse design results for the foam heat transfer problem. **a** Evolution of effective thermal conductivity k_{eff} and sampled structure in the task of designing foam with target effective thermal conductivity k_{eff}^* . The k_{eff} is evaluated based on the structural estimate $\hat{\mathbf{x}}_0$ in each sampling step. **b** Five samples generated with a target effective thermal conductivity of $30 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$. The effective thermal conductivity of each sample is shown below it in $\text{W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$.



model is:

$$d\mathbf{x} = \left[-\frac{1}{2}\beta(t)\mathbf{x} - \beta(t) \left(-\frac{\epsilon_\theta(\mathbf{x}_t, t)}{\sigma_t} - s \nabla_{\mathbf{x}_t} \underbrace{\|k_{\text{eff}}^* - k_{\text{eff}}(\hat{\mathbf{x}}_0(\mathbf{x}_t))\|^2}_{\text{Physics guidance}} \right) \right] dt + \sqrt{\beta(t)} d\mathbf{w} \quad (1)$$

where k_{eff}^* is the target effective thermal conductivity, and $k_{\text{eff}}(\cdot)$ is the physics-based solver used to calculate the effective thermal conductivity of the generated structures. Gradients are computed via customized reverse-mode automatic differentiation (AD), where specialized vector-Jacobian product (VJP) rules are derived from the implicit function theorem²⁵.

The convergence history of k_{eff} toward a target effective thermal conductivity is provided in Fig. 3a. Several intermediate configurations are embedded along the curve, visually demonstrating the structural evolution from random noise to a clean and distinct disordered foam structure as generation progresses. The sampled foam initially shows large geometric variability and significant conductivity fluctuations. As sampling proceeds, dominant heat-transfer paths emerge, and the structures begin to undergo only minor adjustments. Eventually, both the topology and the effective thermal conductivity stabilize around the target value. Five generated foams with a target effective thermal conductivity $k_{\text{eff}}^* = 30 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$ are shown in Fig. 3b. The mean and standard deviation of the k_{eff} of these foams are $30.074 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$ and $0.095 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$, respectively, and the 95% confidence interval (CI) is $[29.957, 30.191] \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$. These values show

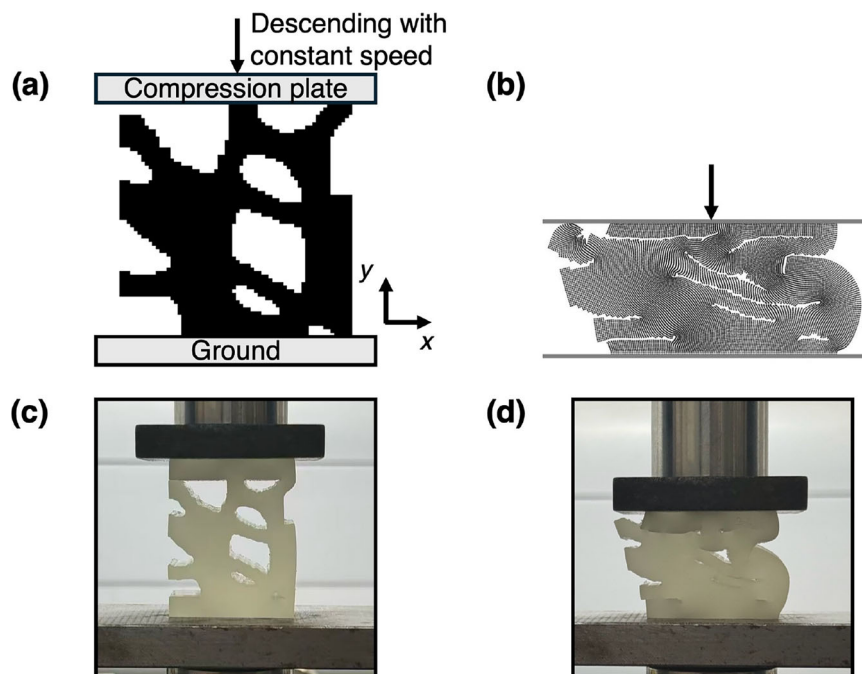
that the generation of foams with target effective thermal conductivity is highly reproducible, with minor sample-to-sample variation. Under the framework of a measurement model $k_{\text{eff}}^* = k_{\text{eff}}(\mathbf{x}_0) + n$, these five independently generated samples exhibit small mean and variance of residuals, which are $-0.074 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$ and $7.1 \times 10^{-3} (\text{W} \cdot \text{m}^{-1} \cdot \text{K}^{-1})^2$, respectively, corresponding to a high likelihood $p(k_{\text{eff}}^* | \mathbf{x}_0)$ and indicating reliable property control across the physics-guided generations.

These results demonstrate the efficacy and accuracy of the physics-guided sampling strategy in generating disordered foams with prescribed effective thermal conductivity.

Control of load-displacement response

We consider the compression of viscoelastic foams in this subsection, with the aim of designing foam structures that exhibit target load-displacement curves. The material considered in this task is flexible photosensitive resin. As shown in Fig. 4a, a foam sample is fixed on the ground, while a top plate moves downward at a constant velocity v_{comp} . The explicit moving least squares material point method (MLS-MPM)²⁶ is applied to simulate this problem. We implement MLS-MPM in JAX²⁷ so that the solver is differentiable. The material behavior is described by a finite-strain viscoelastic constitutive model composed of an equilibrium (hyperelastic) Neo-Hookean response and a non-equilibrium viscous overstress described by a generalized Maxwell model (Prony series)^{28,29}. The damage-field gradient (DFG) partitioning frictional self-contact algorithm is additionally incorporated into our implementation^{30,31}, enabling the automatic detection of potential contact surfaces and effectively preventing the merging of particles

Fig. 4 | Formulation and experimental validation of the foam compression problem. **a** Formulation of the foam compression problem. A top compression plate moves downward at a constant velocity. **b** Compressed state in simulation. **c** Initial state in experiment. **d** Compressed state in experiment.



during the MPM simulation process. The final state of the foam in simulation is depicted in Fig. 4b. More details of the MLS-MPM, constitutive model and frictional self-contact algorithm can be found in Supplementary Section 2 and 3.

The reaction force on the top plate is computed from the internal forces of top boundary nodes, which is given as $F = -\sum_{i \in N_{top}} \mathbf{f}_i \cdot \mathbf{e}_y$, where N_{top} is the set of top boundary nodes, and $\mathbf{f}_i$ is the internal force of grid node i . The load-displacement curve plots load against displacement u . Experiments are performed to calibrate our simulation as shown in Fig. 4c, d, corresponding to the initial and compressed states of the foam resin, respectively. A 3D printer for photosensitive resin is used to fabricate the foams. To facilitate the realization of no-slip boundary conditions on both the ground and the compression plate in the experiments, additional resin layers are applied to the top and bottom surfaces of the foam structure. These resin pads are also incorporated in our simulations to ensure consistency between the experimental setup and the numerical model.

To design foam structures with desired load-displacement curves, we use $n = 100$ discretized points to describe the curve, and evaluate the match degree between the target curve and the current curve of $\hat{\mathbf{x}}_0$ with mean square error (MSE). The physics-guided sampling equation is given as:

$$dx = \left[-\frac{1}{2}\beta(t)\mathbf{x} - \beta(t) \left(-\frac{e_\theta(\mathbf{x}_t, t)}{\sigma_t} - s \nabla_{\mathbf{x}_t} \underbrace{\frac{1}{n} \sum_{j=1}^n \|c_j^* - c_j(\hat{\mathbf{x}}_0(\mathbf{x}_t))\|^2}_{\text{Physics guidance}} \right) \right] dt + \sqrt{\beta(t)} d\mathbf{w} \quad (2)$$

where c^* is the target load-displacement curve, $c(\cdot)$ denotes the physics-based solver to simulate the compression response of given structures, and j is the index of points. In this task, c_j is equal to the load value F_j . Gradients are computed by backpropagation through the explicit MPM time-stepping scheme using the reverse-mode AD.

The evolution of MSE of the load-displacement response during the sampling process is plotted in Fig. 5a. In the early stage, the sampled foam exhibits large structural variations and the force-displacement response differs substantially from the target, resulting in a high MSE. As sampling proceeds, the overall topology becomes more stable and only

minor structural refinements remain. Correspondingly, the force-displacement curve progressively approach the target, and the MSE rapidly decreases to a small level. To systematically evaluate the inverse design capability of the proposed framework across a broad range of mechanical behaviors, four representative target load-displacement curves are designed, as shown in Fig. 5b–e (red lines): (b) hyperelastic behavior, reaching 1000 N at 15 mm displacement; (c) linear response with a maximum load of 500 N; (d) classical foam response³² featuring a pronounced plateau at 100 N between 5 and 10 mm; and (e) complex non-monotonic behavior with extreme buckling and densification. For each case, we generate five different foams and plot the simulation results in Fig. 5b–e (black and blue lines, where the black lines represent the best-matching simulation curves). The root mean square percentage errors (RMSPEs) of these four tasks are 8.17%, 8.14%, 14.93% and 14.00%, respectively.

For all samples generated in this task, we also conduct corresponding compression experiments to obtain the experimental curves. The means and 95% CIs of the simulated and experimental load-displacement curves in each case are plotted in Fig. 6. Notably, all five generated samples in each case demonstrate good agreement with the target curves (red lines), as evidenced by the narrow CIs (blue and green bands) that closely envelope the targets across the entire displacement range. For the simple target curves of cases (a) and (b) shown in Fig. 6, the generated foams exhibit excellent matching with minimal deviations, with both simulation and experimental results tightly clustered around the targets. For the more complex target curves in cases (c) and (d), the matching accuracy is slightly reduced compared to cases (a) and (b). This reduction in numerical accuracy may be attributed to three reasons. First, the target responses are idealized curves, in which the densification plateau and buckling regions are fully flattened or smoothed, whereas fluctuations in the simulated responses may hinder full convergence. Second, the physics-guidance term is defined as the averaged load difference over uniformly selected points on the response curves, which may weaken the influence of the densification and buckling regions; more specialized guidance terms could improve performance by assigning larger weights to non-monotonic responses. Third, non-monotonic responses may be rare or even absent from the training distribution, making it difficult for the diffusion model to generate an ideal design. The experimental errors

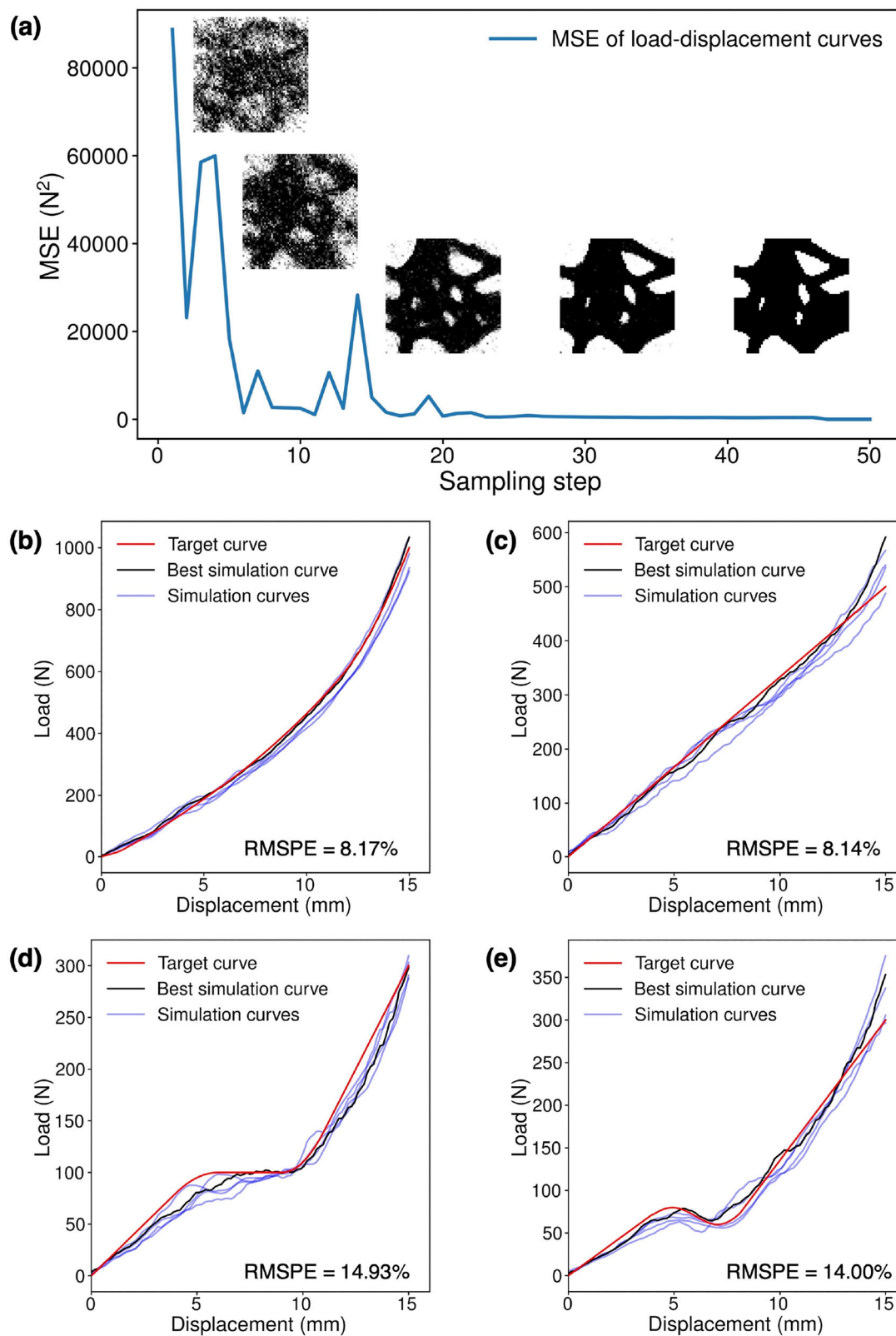


Fig. 5 | Inverse design results for the foam compression problem. **a** Evolution of response MSE and sampled structure in the task of controlling the load-displacement curve. **b–e** Target and simulated load-displacement curves for each

design case, with the best-matching simulation curves highlighted. The root mean square percentage errors (RMSPEs) of each group of curves are also provided within the plots.

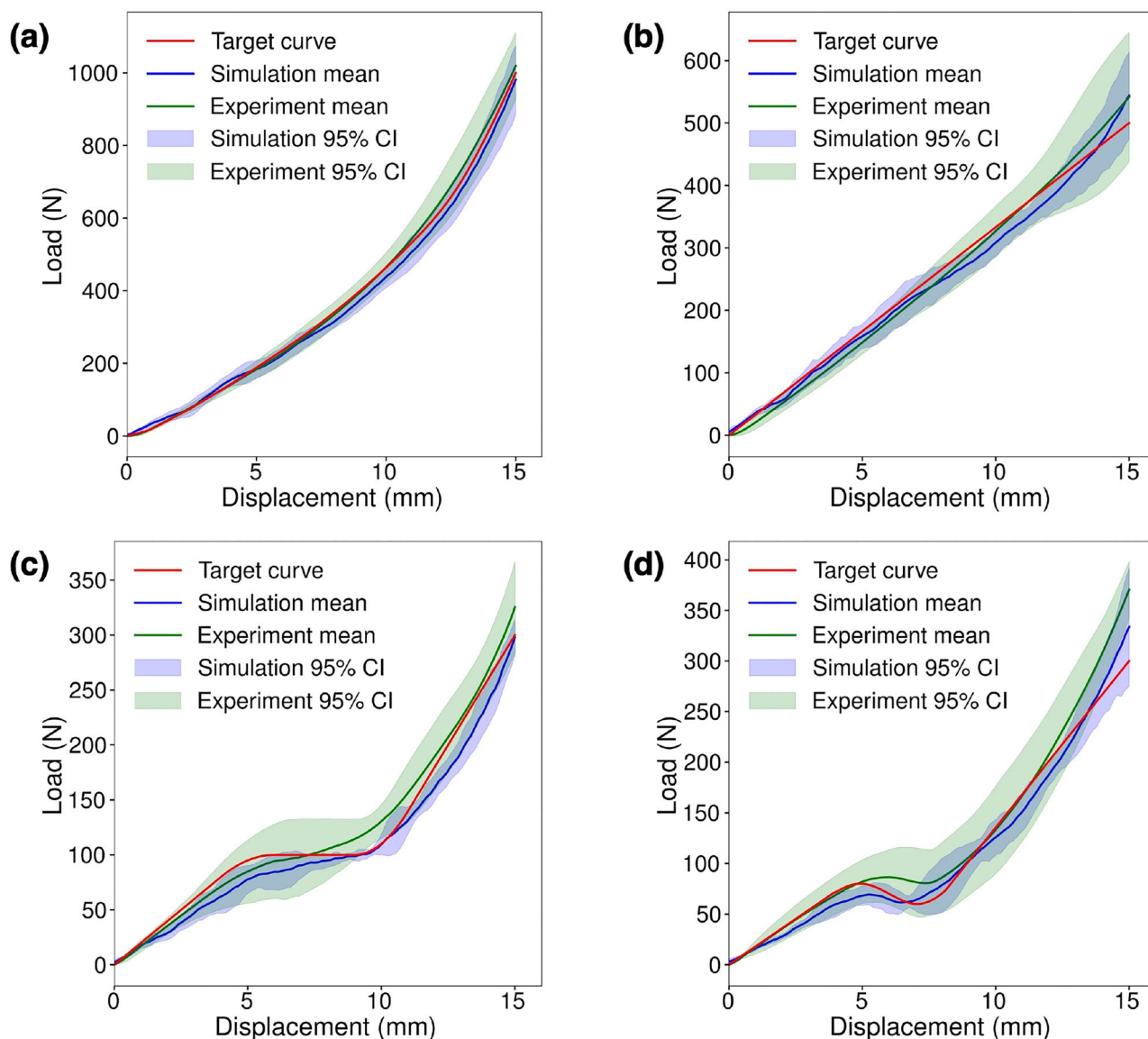


Fig. 6 | Means and 95% confidence intervals (CIs) of the simulated and experimental load-displacement curves of the five generated samples under each design target. The narrow CIs (blue and green bands) that closely encircle the targets over

the whole displacement range show that all five generated samples in each case are in good agreement with the target curves (red lines).

are even larger because the foams were not printed in the same batch. Samples from different batches may exhibit different mechanical responses due to minor variations in printing materials, post-processing procedures, and testing conditions. Nevertheless, the overall trends of all generated samples remain consistent with the targets, successfully capturing characteristic features such as plateau stress levels and densification onset. Future work may further improve the matching accuracy through advances in modeling, including enhanced solver capability, feature-aware guidance terms, and more diverse training samples, as well as through reduced manufacturing variability in experimental validation.

Overall, these results demonstrate that the proposed physics-guided sampling method can reliably generate diverse structures that consistently satisfy the prescribed mechanical specifications, including challenging non-monotonic responses.

Maximization of energy absorption involving fractures

The energy absorption capability of foams enduring fracture is studied in this subsection, and the aim is to design foam structures with maximal absorbed energy given the specified volume fraction. As shown in Fig. 7a,

a foam is fixed on the ground and subjected to a prescribed displacement, and the fracture behavior of the foam is analyzed. This problem is formulated within the phase field brittle fracture framework^{33,34}, where the fracture evolution can be characterized through minimization of the total energy functional with respect to the displacement field $\mathbf{u}$ and the phase field variable ζ . More details of the phase field model can be found in Supplementary Section 4. JAX-FEM²³ is used here to solve this problem, and an example solution of the phase field variable and load-displacement curve is shown in Fig. 7b, c. The absorbed energy is defined as the area under the load-displacement response up to a displacement of 0.01 mm.

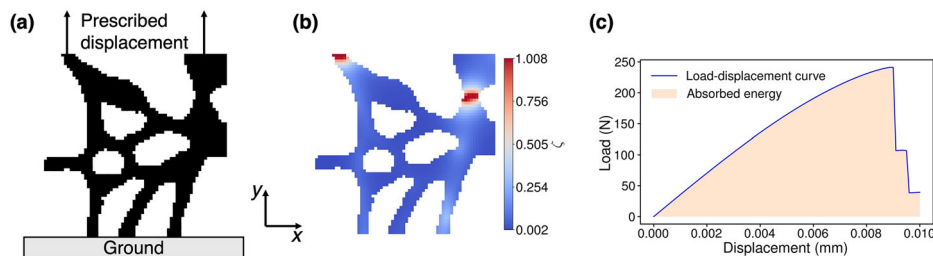
The probability $p(E|\hat{\mathbf{x}}_0)$ is modeled by an energy-based model (EBM) through Boltzmann distribution^{35,36}:

$$p(E|\hat{\mathbf{x}}_0) \propto \exp(-E(\hat{\mathbf{x}}_0)) \quad (3)$$

For the task of maximizing the absorbed energy of generated foams subject to a volume-fraction constraint, the physics-guided sampling equation is expressed as:

$$d\mathbf{x} = \left\{ -\frac{1}{2}\beta(t)\mathbf{x} - \beta(t) \left[-\frac{\epsilon_\theta(\mathbf{x}_t, t)}{\sigma_t} + s \underbrace{\left(-m\nabla_{\mathbf{x}_t} \|\phi^* - \phi(\hat{\mathbf{x}}_0(\mathbf{x}_t))\|^2 + \nabla_{\mathbf{x}_t} E(\hat{\mathbf{x}}_0(\mathbf{x}_t)) \right)}_{\text{Physics guidance}} \right] \right\} dt + \sqrt{\beta(t)} d\mathbf{w} \quad (4)$$

Fig. 7 | Formulation of the foam brittle fracture problem. **a** Formulation of the foam brittle fracture problem. A prescribed displacement is applied on the top boundary. **b** Solution of the phase field variable ζ . **c** Load-displacement curve and absorbed energy.



where ϕ^* is the limitation of volume fraction, $\phi(\cdot)$ is the solver for detecting the volume fraction of the input structure, $E(\cdot)$ is the solver for absorbed energy, and m is a scaling factor. There are two reasons for formulating multi-objective tasks in this way. First, the two guidance terms have different magnitudes, and a scaling factor is therefore needed to balance their contributions. Second, for problems involving maximization or minimization objectives, the scaling factor may require minor adjustment to intentionally break this balance according to the biased target. Otherwise, the optimization may converge to a region far from the Pareto front once the objectives become numerically balanced (e.g., one positive and another negative). By using such a biased scaling factor, the model is encouraged to move away from locally converged regions, explore the design space more thoroughly, and drive the solutions closer to the Pareto front. However, in inverse design problems with multiple exact targets, where no maximization objective is involved, and the loss terms are expected to remain strictly balanced, tuning m may be unnecessary because the individual losses can be directly normalized.

To determine an appropriate value of m for this task, we first compare the magnitudes of the gradients associated with the two design objectives, $\nabla_{\mathbf{x}_t} \|\phi^* - \phi(\hat{\mathbf{x}}_0(\mathbf{x}_t))\|^2$ and $\nabla_{\mathbf{x}_t} E(\hat{\mathbf{x}}_0(\mathbf{x}_t))$, and choose an initial value of m that balances them. Empirically, we find that the volume-guidance term should be slightly stronger than the energy-guidance term to ensure that the generated structures satisfy the volume-fraction constraint. We therefore add a small positive offset to m to intentionally break this balance. Accordingly, the scaling factor is set to $m = 5 \times 10^{-5}$ in this task.

We first randomly generate 10 foam structures without any guidance and evaluate their fracture energies via phase field simulations. Across these samples, the mean volume fraction is 0.442 and the mean absorbed energy is 0.882×10^{-3} J. We then set the volume-fraction constraint ϕ^* to the mean value obtained from these ten structures and perform guided sampling according to Eq. (4).

In Fig. 8a, we plot the evolution of the absorbed energy and the volume fraction along the sampling trajectory, together with the corresponding generated structures. The sampled foam first evolves rapidly to satisfy the prescribed volume fraction, with substantial structural changes in the early stage. After that, the structure undergoes further refinement, leading to a gradual increase and eventual stabilization of the absorbed energy. In Fig. 8b, we present the load-displacement curves and structures of foams generated with and without energy guidance, while the volume-fraction constraint is kept the same. Without energy guidance, the absorbed energy of the generated structure is 0.824×10^{-3} J, while with energy guidance, it increases by 166% to 2.188×10^{-3} J. Five foam structures under the same guidance of limiting the volume fraction and maximizing absorbed energy are shown in Fig. 8c. The mean and standard deviation of the volume fraction across all five samples are 0.452 and 0.005, respectively, and the corresponding 95% CI

is [0.446, 0.458], which closely matches the volume-fraction constraint. The mean and standard deviation of the absorbed energy across these samples are 1.989×10^{-3} J and 0.129×10^{-3} J, respectively, with the 95% CI of $[1.8287, 2.1493] \times 10^{-3}$ J, suggesting that the guided absorbed energy is consistently higher than that of the unguided samples. The guided samples tend to exhibit slightly larger volume fractions than the volume limitation, because adding material is encouraged by the energy-guidance term but penalized by the volume constraint. The resulting designs therefore settle at an intermediate compromise between these competing objectives.

These results demonstrate that the proposed physics-guided sampling strategy can generate foams with high energy absorption performance while satisfying a prescribed volume-fraction constraint. Moreover, the results validate the capability of our method to address multi-constraint inverse design problems.

Comparison of computational cost and accuracy

The computational cost consists of two parts: neural network training and guided foam-structure generation. All experiments are conducted on a MacBook Pro equipped with an Apple M3 Pro chip. Training the network on 6400 binary foam images of size 64×64 takes around 7 h. During structure generation, the differentiable solver is called 50 times for each foam sample and accounts for most of the generation time. As a result, the average computational time for generating a foam with targeted thermal conductivity, load-displacement response and energy absorption is about 45 seconds, 2 hours and 18 hours, respectively.

For previous conditional diffusion models, the computational cost mainly arises during preparation of labeled dataset, as there is no solver call in model training and data sampling. If the paired labels were to be generated by a numerical solver, constructing a training set of 6400 samples normally would require 6400 solver calls.

For previous physics-informed diffusion frameworks where the physical residual is evaluated during training, the solver would be called roughly $6400 \times 50 = 3.2 \times 10^5$ times for a 50-epoch training run (assuming that all data were traversed once in each epoch), so the total cost would scale with both dataset size and training epochs. There is no solver call in data preparation and sampling for this type of methods.

Compared with conditional diffusion models and physics-informed diffusion models, our physics-guided diffusion models shift the solver cost from data preparation and model training to data sampling. However, the sampling process only invokes the differentiable solver 50 times for generating a desired foam structure, without any solver call during data preparation and model training. In addition, when the design objective changes, conditional diffusion models and physics-informed diffusion models typically require retraining, whereas our physics-guided diffusion model saves this time by being trained only once.

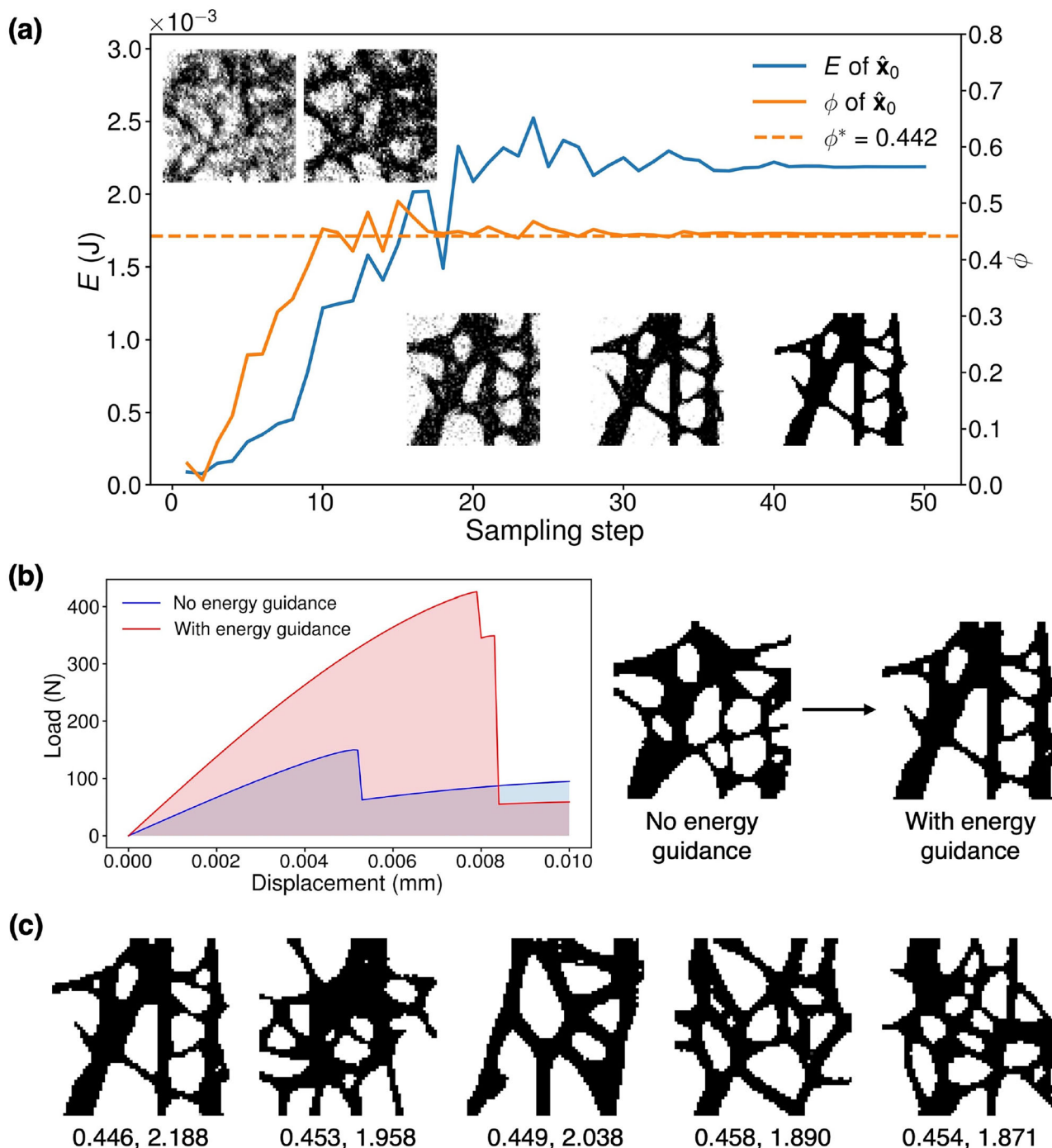


Fig. 8 | Inverse design results for the foam brittle fracture problem. **a** Evolution of absorbed energy E , volume fraction ϕ and sampled structure in the task of maximizing the energy absorption within a volume-fraction constraint ϕ^* . The ϕ and E are evaluated based on the structural estimate $\hat{\mathbf{x}}_0$ in each sampling step. **b** Comparison of load-displacement curves and structures of foams generated with

and without energy guidance, using the same random seed. **c** Five samples generated to maximize energy absorption under a volume-fraction constraint of 0.442. The volume fraction (left) and absorbed energy (right) of each sample are shown below it; the absorbed energy is reported in 10^{-3} J.

To further quantitatively compare the computational cost and accuracy of the physics-guided diffusion model with previous ones, we train a conditional diffusion model and a physics-informed diffusion model for comparison. The task of designing effective thermal conductivity is taken as an example. It should be mentioned that the physics-informed diffusion model is also conditional in this work, as the objective is to generate structures with specified properties. In addition, this physics-informed diffusion model is actually fine-tuned^{37,38} based on the conditional diffusion model for efficiency, with an additional conductivity loss term introduced

into the loss function. We use each model to generate 100 foams and calculate the mean absolute errors (MAEs) of the corresponding effective thermal conductivities. A summary of the three diffusion models is provided in Table 1. The comparative results show that our model requires the least computational time from data preparation to the first sample generation and achieves the highest accuracy compared with the conditional and physics-informed ones.

Since physics guidance is an essential component of the proposed framework, additional effort may be required to develop differentiable

Table 1 | Comparison of computational cost and accuracy for conditional, physics-informed and physics-guided diffusion models

Diffusion model	Computational cost			Accuracy	
	Labeling	Training (each epoch)	Sampling (once)	MAE	Accuracy improvement
Conditional	1037 s	503 s	1.4 s	$17.41 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$	None
Physics-informed	None	1936 s	1.4 s	$8.62 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$	50.49%
Physics-guided (ours)	None	491 s	45.6 s	$0.28 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$	98.39%

The task is to design effective thermal conductivity.

solvers. In practice, applying this framework to domains where forward or differentiable solvers already exist is generally more convenient. However, with the advent of modern automatic differentiation platforms for scientific computing, such as JAX and Taichi, an increasing number of differentiable software packages have become available in recent years, making integration between physics-guided diffusion models and differentiable solvers more flexible.

Generally, our physics-guided diffusion framework is more cost-effective when one aims to (1) save the time for data preparation, (2) keep the training process lightweight, (3) generate a limited number of designs or (4) adapt to different design objectives.

Stability and diversity of generation

The success rate is used to quantitatively evaluate the stability of the proposed physics-guided sampling strategy in generating structures with desired properties. To calculate the success rates, we sampled additional foam structures under different physical targets. All generated samples and their corresponding properties are shown in Fig. 9b–d. For the effective thermal conductivity design task, the acceptable relative error for success is defined as 1%. The framework performs well on this simple task, achieving a success rate of 100%. For the load-displacement control task, the relative error tolerance is defined as 15% since this task requires matching complete response curves, and the corresponding success rate is 80%. Finally, for the task of maximizing energy absorption under a volume constraint, success is defined as satisfying the volume constraint within the relative error tolerance of 5% while achieving an energy absorption greater than twice the average value (see Section Maximization of energy absorption involving fractures). For this task with multiple objectives, the success rate is also 80%. Overall, the success rates across the three tasks investigated in this work are promising, demonstrating that the proposed physics-guided diffusion model can generate target structures with relatively stable performance.

The ability of out-of-distribution generation is also evaluated upon the task of designing the effective thermal conductivity k_{eff} . We first calculate the effective thermal conductivities for all the samples in the training dataset, and find out the range of these k_{eff} is $[12.73, 81.92] \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$. Then we choose several out-of-distribution thermal conductivities, which are 5, 10, 85, 90 and $95 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$, as the design targets k_{eff}^* and perform the sampling. The results are shown in Fig. 10. When the target properties lie slightly outside the training distribution, the model can still generate reasonable samples while largely preserving both morphology and property. However, as the target properties move further away from the training distribution, although physics guidance can still enforce the desired properties, the generated samples gradually deviate from the learned foam prior.

To investigate the sensitivity of model performance to the training dataset size, we conduct an ablation study. Specifically, we train three additional diffusion models using 1600, 400, and 100 images, respectively, and apply them to generate 100 foams under different target effective thermal conductivities. The MAE is used to quantify the discrepancy in physical properties, while the Fréchet inception distance (FID)³⁹ is used to evaluate generation quality. Table 2 summarizes the results, showing that generation quality decreases as the dataset size becomes smaller, whereas physical consistency remains relatively insensitive to dataset size. This outcome is expected, since a smaller dataset makes it more difficult for the

diffusion model to capture sufficient prior knowledge. However, regardless of sample quality, the generation process still remains constrained by physics guidance.

In terms of the diversity of generated samples, we present some samples from the training dataset and additional samples generated by the physics-guided diffusion model in Fig. 9 for visual assessment. Each group of samples exhibits diverse morphologies while maintaining similar physical properties. In addition to being distinct from one another, the generated samples also differ from those in the training dataset. One notable difference is that the foams in the training dataset, which are generated by Voronoi tessellation (see Section Methods), tend to have relatively uniform wall thickness, whereas the generated samples may exhibit more uneven wall thickness. We further perform a quantitative analysis of diversity using the Intersection over Union (IoU)⁴⁰ and latent-space distance metrics, both of which reflect sample diversity. Again, we take the effective thermal conductivity design task as an example. Specifically, we first identify the samples in the training dataset with effective thermal conductivities ranging from 29.5 to $30.5 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$, yielding approximately 75 samples. We then generate 75 foams using the physics-guided diffusion model under a target conductivity of $30 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$ and compute the pairwise similarity and distance statistics. The average IoU between generated samples and training samples is 0.386, with a maximum of 0.649 and a minimum of 0.226, whereas the average IoU among generated samples is 0.404, with a maximum of 0.792 and a minimum of 0.232. The IoU statistics show that the generated samples are not simple replicas of either the training samples or one another. For the latent-space analysis, we train an autoencoder on the full training dataset, project the samples into the latent space, and then calculate pairwise distances. The average pairwise latent-space distance among the 75 training samples is 10.53, with a maximum of 43.77 and a minimum of 0.08, whereas the average pairwise latent-space distance among generated samples is 11.46, with a maximum of 45.49 and a minimum of 0.21. The latent-space distances among generated samples are slightly larger than those among the corresponding training samples, suggesting that the generated set preserves a level of morphological diversity comparable to that of the training data. Besides, we visualize these latent-space projected samples in a 2D scatter plot using the first two principal components (PCs) of them, as shown in Fig. 11. The figure shows that the generated samples are diverse and span a design space comparable to that of the training distribution. Together, these results indicate that the generated samples are highly diverse, not only with respect to one another but also with respect to the training data.

Discussion

We propose physics-guided diffusion models that leverage differentiable physics-based solvers to address inverse design problems of 2D disordered closed-cell foam metamaterials. We begin by training a score-based diffusion model formulated through SDEs. Inspired by the concept of classifier guidance, we further modify the sampling process by incorporating physics-based guidance into the SDEs, where the guidance is evaluated via differentiable physics solvers. To validate the proposed approach, we provide three numerical case studies: (1) the design of foams with target effective thermal conductivity, (2) tailoring the load-displacement response, and (3) maximizing energy absorption under a given volume-fraction constraint.

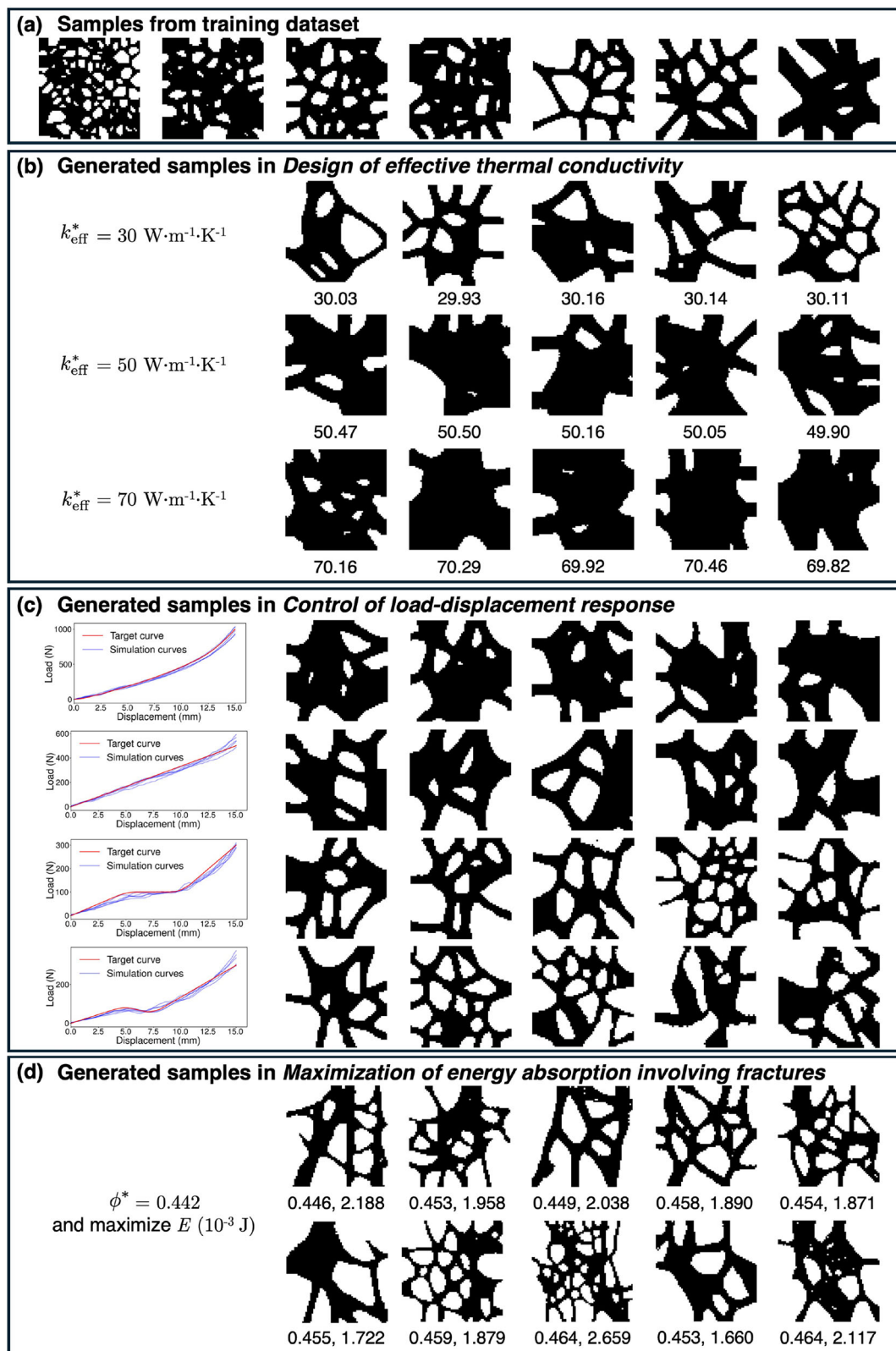


Fig. 9 | Examples of foam samples. a Samples from the training dataset. **b** Samples generated with target effective thermal conductivities k_{eff}^* ; the effective thermal conductivity of each sample is shown below it in $\text{W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$. **c** Samples generated with target mechanical responses, with the simulated load-displacement curve of

each sample plotted. **d** Samples generated to maximize energy absorption E under a volume-fraction constraint of $\phi^* = 0.442$. The volume fraction (left) and absorbed energy (right) of each sample are shown below it; the absorbed energy is reported in 10^{-3} J .

Fig. 10 | Results of out-of-distribution sampling in the design of effective thermal conductivity task. The target effective thermal conductivity k_{eff}^* and effective thermal conductivity k_{eff} of each sample are shown under the generated foams. The results show that slight extrapolation beyond the training distribution remains feasible, but larger deviations lead to samples departing from the learned foam prior.

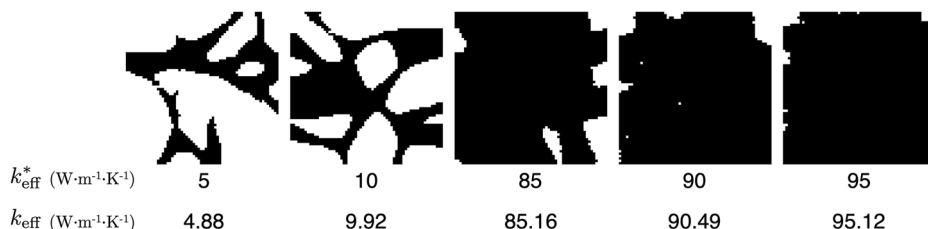


Table 2 | MAEs and FIDs of generated samples from models trained with different dataset sizes

Number of images for training	6400	1600	400	100
MAE ($\text{W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$)	0.28	0.45	0.27	0.33
FID	99.89	113.24	119.34	145.74

The task is to design effective thermal conductivity.

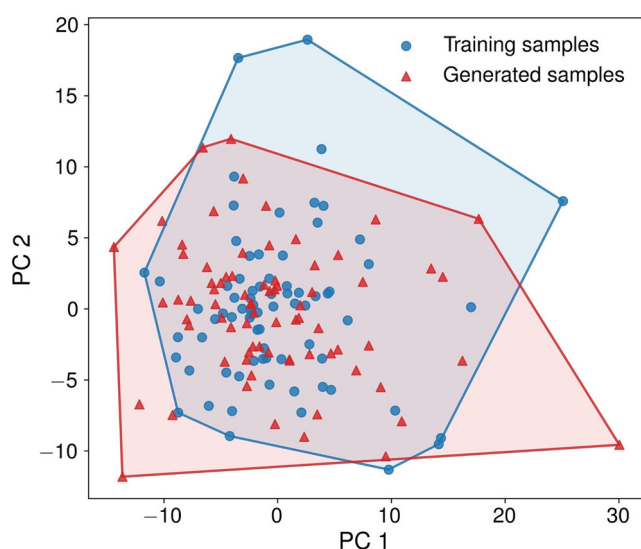


Fig. 11 | Scatter plot of training and generated samples in latent space. The 75 training samples and 75 generated samples with effective thermal conductivities around $30 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$ are projected into a 2D latent space using an autoencoder. The axes represent the first two principal components (PCs) of the projected samples. This figure shows that the generated samples are diverse and cover a design space comparable to that of the training distribution.

The results consistently demonstrate the efficacy of physics-guided diffusion models in solving challenging inverse design tasks.

In contrast to conventional generative models that require labeling in data preparation or physics residuals in model training, the proposed physics-guided diffusion framework exhibits clear advantages in the inverse design of disordered metamaterials. First, this framework guarantees the physics consistency: By integrating differentiable physics solvers into the generative process, the approach enables direct steering of the sampling trajectory towards structures that satisfy desired properties, such as specific thermal conductivity, load-displacement response, or energy absorption, as shown in the diverse case studies in this work. Second, the model supports label-free and solver-decoupled training: The proposed diffusion framework can be trained on unlabeled datasets, avoiding the expensive generation of property labels typically required by conditional generative models. Moreover, by decoupling the physics solver from training and using it only for sampling, the approach avoids repeated solver evaluations and gradient

backpropagation across batches during training, leading to substantially improved computational efficiency and stability. Third, the framework also shows flexibility across multiple design objectives within a learned morphology prior: The physics-guided approach can be flexibly adapted to a variety of inverse design objectives by changing the differentiable solver and modifying the guidance term, as evidenced by its effective performance across all three different design objectives without retraining.

While the proposed framework demonstrates strong capabilities, several directions remain promising for future work. First, broader and more diverse data curation could expand the reachable design space, as the generative process currently explores patterns within the learned morphological prior. Second, the current framework is limited to 2D foams, but the physics-guided generative pipeline should also be applicable to 3D problems in principle⁴¹. Third, automated hyperparameter-calibration strategies would further improve flexibility and robustness, as performance depends on physics-guidance intensity and other scaling factors that can affect convergence and generation quality. Finally, developing faster and higher-fidelity physics solvers or learned surrogates would reduce computational cost for complex or large-scale systems.

Methods

In this section, we first describe the procedure of generating disordered closed-cell foam images via Voronoi tessellation. We then review the classical score-based generative models formulated with SDEs, as well as classifier-guidance diffusion models. Then, we present our physics-guided diffusion models for inverse design.

Voronoi tessellation

The objective of this study is to design disordered closed-cell foam materials. To construct the dataset of disordered foams, we employ the Voronoi tessellation method^{42,43}, which partitions a plane into regions based on distance to a specific set of seed points. We generate 6400 binary images of disordered closed-cell foams at 64×64 resolution for training, with varying cell diameters and volume fractions. Detailed implementation of Voronoi tessellation can be found in Supplementary Section 5.

Score-based generative models

Song et al.¹⁶ unified two successful classes of probabilistic generative models, score matching with Langevin dynamics (SMLD)⁴⁴ and denoising diffusion probabilistic modeling (DDPM)¹⁵, by using score-based generative framework through SDEs. We adopt this score-based generative framework in this work, as the SDE formulation enables flexible and controllable data generation.

Following Song et al.¹⁶, we consider a forward process that perturbs original data $\mathbf{x}(0) \sim p_0$ into $\mathbf{x}(T) \sim p_T$ over a continuous time interval $t \in [0, T]$, where p_T is a Gaussian distribution with fixed mean and variance. This forward diffusion process is described by the following SDE:

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w} \quad (5)$$

where $\mathbf{f}(\cdot, t) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a vector-valued function called the drift coefficient of $\mathbf{x}(t)$, $g(\cdot) : \mathbb{R} \rightarrow \mathbb{R}$ is a scalar function called the diffusion coefficient of $\mathbf{x}(t)$, and $\mathbf{w}$ is a standard Wiener process. In this work, we

employ the variance-preserving (VP) SDE:

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}dt + \sqrt{\beta(t)}d\mathbf{w} \quad (6)$$

where $\beta(t)$ is a monotonically increasing noise schedule function. The VP-SDE has the key property that the marginal distribution $p_t(\mathbf{x}_t|\mathbf{x}_0)$ is analytically tractable:

$$p_t(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}) \quad (7)$$

where $\bar{\alpha}_t = \exp\left(-\int_0^t \beta(s)ds\right)$.

The reverse process is obtaining the samples $\mathbf{x}(0) \sim p_0$ by starting from prior data distribution $\mathbf{x}(T) \sim p_T$, which is also a diffusion process⁴⁵ and can be given by the reverse SDE:

$$d\mathbf{x} = [\mathbf{f}(\mathbf{x}, t) - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x})]dt + g(t)d\bar{\mathbf{w}} \quad (8)$$

where dt now flows backward from T to 0 , $d\bar{\mathbf{w}}$ is a reverse-time Wiener process, and $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ is the score function of the marginal density $p_t(\mathbf{x})$.

The score function $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ is approximated by a time-dependent neural network $\mathbf{s}_\theta(\mathbf{x}, t)$. Given the transition distribution $p_t(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, \sigma_t^2\mathbf{I})$ where $\sigma_t^2 = 1 - \bar{\alpha}_t$, we can write $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sigma_t\epsilon$ with $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The score function admits the following representation:

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) = -\frac{\mathbb{E}[\epsilon|\mathbf{x}_t]}{\sigma_t} = -\frac{\epsilon_\theta(\mathbf{x}_t, t)}{\sigma_t} \quad (9)$$

where $\epsilon_\theta(\mathbf{x}_t, t)$ is a neural network trained to predict the noise ϵ from $\mathbf{x}_t$. This equivalence between score matching and noise prediction enables efficient training via denoising score matching^{15,16}.

Substituting the VP-SDE drift coefficient and the noise-based score representation into the reverse SDE, we can obtain the sampling equation:

$$d\mathbf{x} = \left[-\frac{1}{2}\beta(t)\mathbf{x} + \beta(t)\frac{\epsilon_\theta(\mathbf{x}_t, t)}{\sigma_t}\right]dt + \sqrt{\beta(t)}d\bar{\mathbf{w}} \quad (10)$$

This SDE can be solved numerically using standard SDE solvers from $t = T$ to $t = 0$ to generate samples from the learned data distribution.

A U-Net⁴⁶ architecture is adopted to predict the noise $\epsilon_\theta(\mathbf{x}_t, t)$. The U-Net follows a symmetrical encoder-decoder architecture consisting of four downsampling and four upsampling blocks. Feature channels expand progressively as (64, 128, 256, 512) to encode multi-scale representations. Each block incorporates two residual blocks for adequate feature learning. Self-attention mechanisms are selectively applied at the third and fourth blocks (256 and 512 channels) to balance efficiency and representation power, utilizing 4 attention heads with 32-group normalization. Temporal information is encoded through sinusoidal embeddings and subsequently transformed via a two-layer multilayer perceptron into a 256-dimensional conditioning vector. This time encoding is additively integrated into every residual block, enabling global temporal guidance throughout the network hierarchy. The prepared 6400 binary foam images are used to train this U-Net.

Classifier guidance

The score-based model described above is unconditional, so it can only generate random samples from the learned data distribution $p(\mathbf{x})$. However, for conditional generation tasks, we seek to generate data that have specific attributes. Classifier guidance²¹ provides a principled approach to guide the generative process to desired categories.

The core idea of classifier guidance is to leverage a pre-trained classifier $p_\kappa(y|\mathbf{x})$ to steer the reverse diffusion process toward a target class y^* . By

Bayes' rule, the conditional score $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t|y^*)$ can be decomposed as:

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t|y^*) = \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(y^*|\mathbf{x}_t) \quad (11)$$

where the first term on the right-hand side of this equation is the unconditional score, while the second term $\nabla_{\mathbf{x}_t} \log p(y^*|\mathbf{x}_t)$ represents the gradient of the log-probability that the classifier assigns to label y^* given the input $\mathbf{x}_t$, with respect to $\mathbf{x}_t$.

To apply classifier guidance, the classifier must be trained on noisy samples at various noise levels, so it learns to predict categories from partially perturbed data $\mathbf{x}_t$. The conditional reverse SDE then becomes:

$$d\mathbf{x} = \left[\mathbf{f}(\mathbf{x}, t) - g(t)^2 \left(\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) + s \nabla_{\mathbf{x}_t} \log p_\kappa(y^*|\mathbf{x}_t)\right)\right]dt + g(t)d\bar{\mathbf{w}} \quad (12)$$

where s is a guidance scale parameter that controls how strongly the classifier influences the generation process.

While classifier guidance has proven effective for categorical generation tasks in computer vision²¹, it faces challenges when applied to material inverse design problems. First, training accurate classifiers on noisy intermediate structures is difficult. Second, the classifier introduces approximation errors that can accumulate throughout the iterative generative process, potentially leading to physically inconsistent structures. Third, modifying design specifications or adding new optimization objectives requires retraining the classifier on appropriately labeled data, which is both computationally expensive and labor-intensive. Fourth, many inverse design problems are inherently regression tasks rather than classification problems.

These limitations motivate the development of our physics-guidance sampling strategy in diffusion models, where the classifier is replaced with direct physics-based simulations to provide more accurate and reliable guidance signals.

Physics guidance

To overcome the limitations of classifier guidance, we propose physics-guided diffusion models that directly incorporate physics-based simulations into the generative process. Our objective is to generate 2D disordered closed-cell foam structures $\mathbf{x}$ that achieve specific target physical properties f^* .

Following the conditional sampling framework, we begin with Bayes' rule. The score of the physics-constrained distribution can be decomposed as:

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t|f^*) = \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(f^*|\mathbf{x}_t) \quad (13)$$

where the second term on the right-hand side of the equation represents the gradient of log probability $\log p(f^*|\mathbf{x}_t)$ with respect to the input $\mathbf{x}_t$. Unlike classifier guidance which relies on a learned surrogate model, we aim to construct this second term directly from physics simulations.

The ultimate goal in this work is to obtain a clean, noise-free image that can be fed into the physics solver $f(\cdot)$ to produce accurate results. However, during the reverse diffusion process, we only have access to noisy intermediate states $\mathbf{x}_t$, which can introduce numerical errors and may even cause the solver to fail to converge. To directly optimize the final state $\mathbf{x}_0$, we use the estimation of clean data from the noisy observation to perform the physics-based calculations, which is given as⁴⁷:

$$\hat{\mathbf{x}}_0 = \mathbb{E}[\mathbf{x}_0|\mathbf{x}_t] = \frac{\mathbf{x}_t - \sigma_t \epsilon_\theta(\mathbf{x}_t, t)}{\sqrt{\bar{\alpha}_t}} \quad (14)$$

This predicted $\hat{\mathbf{x}}_0$ serves as a differentiable proxy for the clean structure, enabling us to evaluate physics-based objectives during the noisy sampling

process. By the law of total probability, we have:

$$p(f^*|\mathbf{x}_t) = \int p(f^*|\mathbf{x}_0) p(\mathbf{x}_0|\mathbf{x}_t) d\mathbf{x}_0 \quad (15)$$

where $p(\mathbf{x}_0|\mathbf{x}_t)$ can be approximated by a Dirac delta function $\delta(\mathbf{x}_0 - \hat{\mathbf{x}}_0)$, as $\hat{\mathbf{x}}_0$ is the estimated mean of $p(\mathbf{x}_0|\mathbf{x}_t)$. Then Eq. (15) can be further written as:

$$p(f^*|\mathbf{x}_t) \approx \int p(f^*|\mathbf{x}_0) \delta(\mathbf{x}_0 - \hat{\mathbf{x}}_0) d\mathbf{x}_0 = p(f^*|\hat{\mathbf{x}}_0) \quad (16)$$

The approximation $p(f^*|\mathbf{x}_t) \approx p(f^*|\hat{\mathbf{x}}_0)$ becomes increasingly accurate as the posterior $p(\mathbf{x}_0|\mathbf{x}_t)$ concentrates during later stages of the reverse process (small t). While the approximation is less precise at early stages (large t), it empirically provides effective guidance throughout sampling⁴⁸. Considering a measurement model $f^* = f(\hat{\mathbf{x}}_0) + n$ with $n \sim \mathcal{N}(0, \tau)$, we have:

$$p(f^*|\hat{\mathbf{x}}_0) = \frac{1}{\sqrt{2\pi\tau}} \exp\left[-\frac{\|f^* - f(\hat{\mathbf{x}}_0)\|^2}{2\tau}\right] \quad (17)$$

Then the second term on the right-hand side of Eq. (13) can be written as:

$$\nabla_{\mathbf{x}_t} \log p(f^*|\mathbf{x}_t) = -\frac{1}{2\tau} \nabla_{\mathbf{x}_t} \|f^* - f(\hat{\mathbf{x}}_0(\mathbf{x}_t))\|^2 \quad (18)$$

Combining with the unconditional and classifier-guided reverse SDEs (Eq. (10) and (12)), we obtain the physics-guided sampling SDE for the VP diffusion model:

$$d\mathbf{x} = \left[-\frac{1}{2}\beta(t)\mathbf{x} - \beta(t) \left(-\frac{\epsilon_\theta(\mathbf{x}_t, t)}{\sigma_t} - s \underbrace{\nabla_{\mathbf{x}_t} \|f^* - f(\hat{\mathbf{x}}_0(\mathbf{x}_t))\|^2}_{\text{Physics guidance}} \right) \right] dt + \sqrt{\beta(t)} d\mathbf{w} \quad (19)$$

where $s = 1/2\tau$, which can be regarded as a coefficient controlling the intensity of guidance. In practice, we normalize the guidance term to ensure numerical stability during sampling.

Alternatively, we can apply the chain rule to transform the gradient from $\mathbf{x}_t$ space to $\hat{\mathbf{x}}_0$ space:

$$\nabla_{\mathbf{x}_t} \log p(f^*|\mathbf{x}_t) = -s \frac{\partial \hat{\mathbf{x}}_0(\mathbf{x}_t)}{\partial \mathbf{x}_t} \nabla_{\hat{\mathbf{x}}_0} \|f^* - f(\hat{\mathbf{x}}_0(\mathbf{x}_t))\|^2 \quad (20)$$

Then the physics-guided sampling equation can be expressed as:

$$d\mathbf{x} = \left[-\frac{1}{2}\beta(t)\mathbf{x} - \beta(t) \left(-\frac{\epsilon_\theta(\mathbf{x}_t, t)}{\sigma_t} - s \frac{\partial \hat{\mathbf{x}}_0}{\partial \mathbf{x}_t} \nabla_{\hat{\mathbf{x}}_0} \|f^* - f(\hat{\mathbf{x}}_0(\mathbf{x}_t))\|^2 \right) \right] dt + \sqrt{\beta(t)} d\mathbf{w} \quad (21)$$

where the Jacobian $\partial \hat{\mathbf{x}}_0 / \partial \mathbf{x}_t$ is given as:

$$\frac{\partial \hat{\mathbf{x}}_0}{\partial \mathbf{x}_t} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{I} - \sigma_t \frac{\partial \epsilon_\theta(\mathbf{x}_t, t)}{\partial \mathbf{x}_t} \right) \quad (22)$$

The U-Net training, physics-based simulations and gradient computations in this study are all implemented with JAX²⁷, a high-performance numerical computing library with built-in GPU acceleration and AD capabilities, to ensure seamless integration of physics solvers with the diffusion model and maintain computational efficiency throughout the physics-guided sampling process. In particular, the gradient $\nabla_{\mathbf{x}_t} \|f^* - f(\hat{\mathbf{x}}_0(\mathbf{x}_t))\|^2$ is provided directly by differentiable physics simulators. As higher-fidelity physical models become available, our framework can leverage them to

Table 3 | Parameters used in the physics-guided diffusion model

Parameters	value
Image resolution	64
Channel	2
Batch size	32
Learning rate	2×10^{-4}
Epochs	50
Sampling steps	50
t	[0, 1]
$\beta(t)$	$20t$
$\eta(t)$	$5 \times 2^{1-t}$
$s(t)$	$50(1 - t)$

tackle increasingly complex physics, without being tied to any specific solver. In the following section, we demonstrate this advantage through a set of numerical case studies using differentiable FEM or differentiable MPM.

Projection and model parameters

The estimation $\hat{\mathbf{x}}_0$ is often noisy, particularly in the early iterations of the reverse diffusion process. Since the physics solvers require binary foam images and may fail to converge on highly noisy inputs, we rescale $\hat{\mathbf{x}}_0$ to [0, 1] and subsequently apply a projection to obtain a near-binary image. The projection function $\mathcal{P}$ is defined as:

$$\mathcal{P}(\hat{\mathbf{x}}_0) = \frac{\tanh(\frac{\eta}{2}) + \tanh[\eta(\hat{\mathbf{x}}_0 - \frac{1}{2})]}{2 \tanh(\frac{\eta}{2})} \quad (23)$$

where η is a hyperparameter that controls the sharpness of the projection. A small η results in a smooth transition between phases, while a large η yields a more binary distribution. At the beginning of sampling, η is set to a relatively small value to allow the model to freely explore the design space. As sampling proceeds, we gradually increase η so that the final generated foam structure becomes fully binary. The material inherent property ξ for each phase is then assigned according to a linear interpolation scheme:

$$\xi(\hat{\mathbf{x}}_0) = \xi_0 + (\xi_1 - \xi_0)\mathcal{P}(\hat{\mathbf{x}}_0) \quad (24)$$

where ξ_0 and ξ_1 represent the inherent properties of two phases, and ξ is the material property fed into the physics-based simulators during the sampling process.

The hyperparameter s in Eq. (19) which controls the guidance intensity also needs to be carefully designed. Similar to the η schedule, we use a low guidance intensity at the beginning of sampling to encourage exploration, and gradually increase it toward the end to strictly steer the generated structure toward the target properties.

The parameters used in our physics-guided diffusion model are listed in Table 3 for reference. Since the hyperparameters listed in this table are kept the same across all three independent tasks considered in this paper, we recommend using this table to initialize the hyperparameters when applying the proposed framework to other inverse-design scenarios.

Data availability

All data generated or analyzed during this study are included in this article and its supplementary information files. The underlying code for this study is available at <https://github.com/CMSL-HKUST/physics-guidance>.

Code availability

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Received: 4 June 2026; Accepted: 3 September 2026;

Published online: 17 September 2026

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Acknowledgements

This work was financially supported by National Natural Science Foundation of China (YSF 1240020027, JRS N_HKUST687/25), Research Grants Council of Hong Kong (ECS 26205024, CRF C7085-24G) and Innovation and Technology Fund (ITS/032/23).

Author contributions

Z.X. contributed to conceptualization, methodology, software, investigation, and writing the original draft. W.X. contributed to software, investigation, writing, review, and editing. D.Z. contributed to software, investigation, writing, review, and editing. W.Z., D.D., B.X., and N.L. contributed to investigation. S.M. contributed to writing, review, and editing. T.X. contributed to conceptualization, methodology, writing, review and editing, supervision, and funding acquisition. All authors read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41524-026-02328-y>.

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